

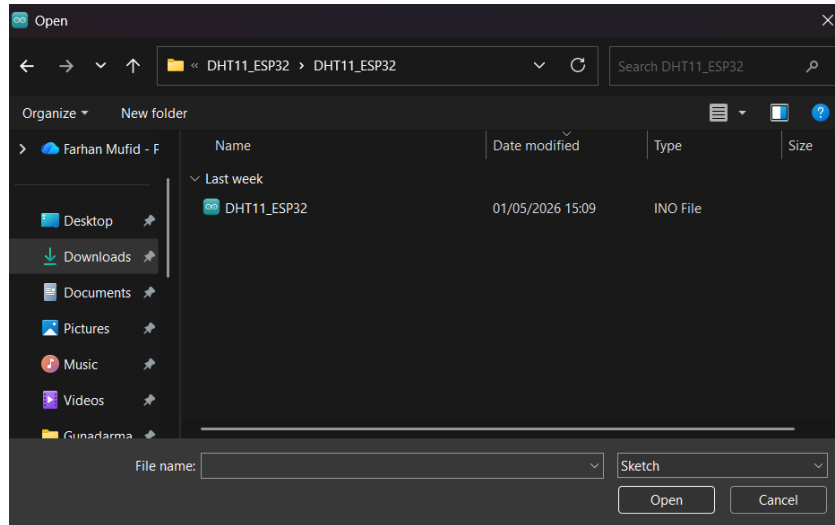
ARDUINO

1. Install App Arduino IDE

2. Install DHT11_ESP32.rar pada link ini dan extract

<https://drive.google.com/file/d/1VCpT-Kl0thh3YK9aL6taunYHEctkBR3N/view>

3. Buka File tersebut pada app Arduino



4. Ubah pada bagian DHT11_ESP32.ino menjadi kodingan ini

```
// 1. DEFINISI BLYNK (wajib di urutan paling atas)

#define BLYNK_TEMPLATE_ID "TMPL6TGzIZmu5"
#define BLYNK_TEMPLATE_NAME "Monitoring DHT 11"
#define BLYNK_DEVICE_NAME "Monitoring DHT 11"
#define BLYNK_FIRMWARE_VERSION "0.1.0"

#define BLYNK_PRINT Serial
#define APP_DEBUG

// 2. PEMANGGILAN LIBRARY
#include "DHT.h"
#include "BlynkEdgent.h"

// 3. PENGATURAN SENSOR DHT11
#define DHTPIN 23          // Kabel DATA DHT11 terhubung ke pin 23
#define DHTTYPE DHT11      // Jenis sensor

DHT dht(DHTPIN, DHTTYPE);
BlynkTimer timer;         // Inisialisasi timer Blynk

// 4. FUNGSI PEMBACAAN DAN PENGIRIMAN DATA KE BLYNK
```

```

void sendSensor() {
    float h = dht.readHumidity();
    float t = dht.readTemperature(); // Membaca suhu dalam Celcius

    // Cek jika sensor gagal terbaca, lalu batalkan pengiriman
    if (isnan(h) || isnan(t)) {
        Serial.println("Gagal membaca sensor DHT!");
        return;
    }

    // Kirim data ke Virtual Pin Blynk sesuai pengaturan Datastream
    Blynk.virtualWrite(V0, t); // Kirim nilai Suhu ke Virtual Pin V0
    Blynk.virtualWrite(V1, h); // Kirim nilai Kelembapan ke Virtual Pin V1

    // Tampilkan data di Serial Monitor untuk pengecekan lokal
    Serial.print("Suhu: ");
    Serial.print(t);
    Serial.print(" °C | Kelembapan: ");
    Serial.print(h);
    Serial.println(" %");
}

// 5. FUNGSI SETUP (Berjalan satu kali)
void setup() {
    Serial.begin(115200);
    delay(100);

    dht.begin(); // Inisialisasi sensor DHT
    BlynkEdgent.begin(); // Inisialisasi koneksi Blynk dan WiFi

    // Atur interval pengiriman data ke server setiap 2000 milidetik (2 detik)
    timer.setInterval(2000L, sendSensor);
}

// 6. FUNGSI LOOP (Berjalan terus-menerus)
void loop() {
    BlynkEdgent.run(); // Menjaga koneksi ke server Blynk tetap hidup
    timer.run(); // Mengeksekusi fungsi sendSensor sesuai interval waktu
}

```

5. Lalu Ubah pada bagian Console.h menjadi kodingan program seperti ini

```
#include <Blynk/BlynkConsole.h>
```

```

BlynkConsole    edgentConsole;

void console_init()
{
    edgentConsole.begin(BLYNK_PRINT);

    edgentConsole.print("\n>");

    edgentConsole.addCommand("reboot", []() {
        edgentConsole.print(R"json({"status":"OK","msg":"resetting device"})json"
"\n");
        delay(100);
        restartMCU();
    });

    edgentConsole.addCommand("config", []() {
        edgentConsole.print(R"json({"status":"OK","msg":"entering configuration
mode"})json" "\n");
        BlynkState::set(MODE_WAIT_CONFIG);
    });

    edgentConsole.addCommand("devinfo", []() {
        edgentConsole.printf(
            R"json({"board":"%s","tpl_id":"%s","fw_type":"%s","fw_ver":"%s"})json"
"\n",
            BLYNK_DEVICE_NAME,
            BLYNK_TEMPLATE_ID,
            BLYNK_FIRMWARE_TYPE,
            BLYNK_FIRMWARE_VERSION
        );
    });

    edgentConsole.addCommand("netinfo", []() {
        char ssidBuff[64];
        getWiFiName(ssidBuff, sizeof(ssidBuff));

        byte mac[6] = { 0, };
        WiFi.macAddress(mac);

        edgentConsole.printf(
            R"json({"ssid":"%s","bssid":"%02x:%02x:%02x:%02x:%02x:%02x","rssi":%d})js
on" "\n",
            ssidBuff,
            mac[5], mac[4], mac[3], mac[2], mac[1], mac[0],
            WiFi.RSSI()

```

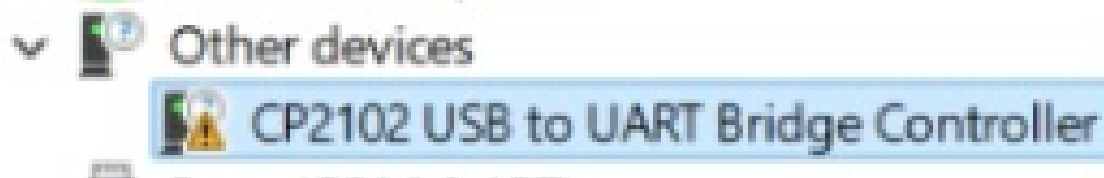
```

    );
  });
}

BLYNK_WRITE(InternalPinDBG) {
  String cmd = String(param.asStr()) + "\n";
  edgntConsole.runCommand((char*)cmd.c_str());
}

```

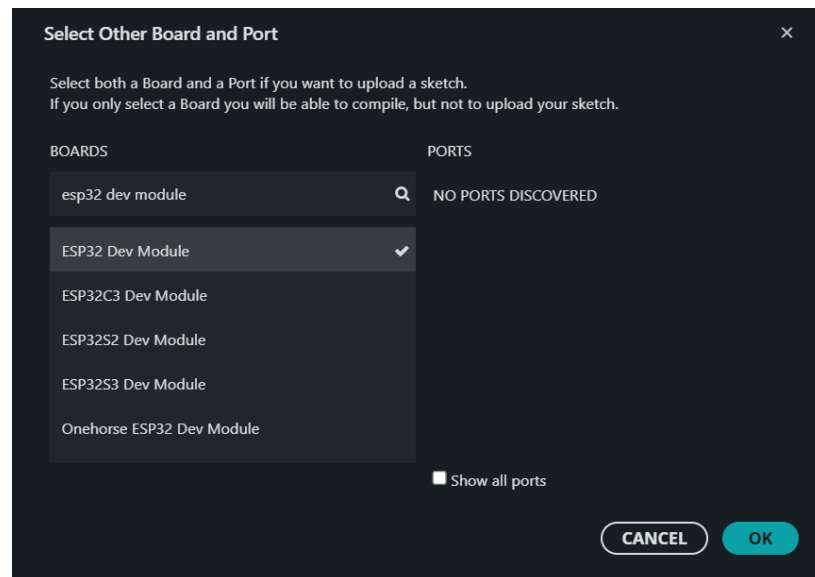
6. Pasang alat lot pada laptop, dan jangan lupa untuk update windows/ update drive port (case missal pada device manager ada contoh port dengan gambar tanda seru)



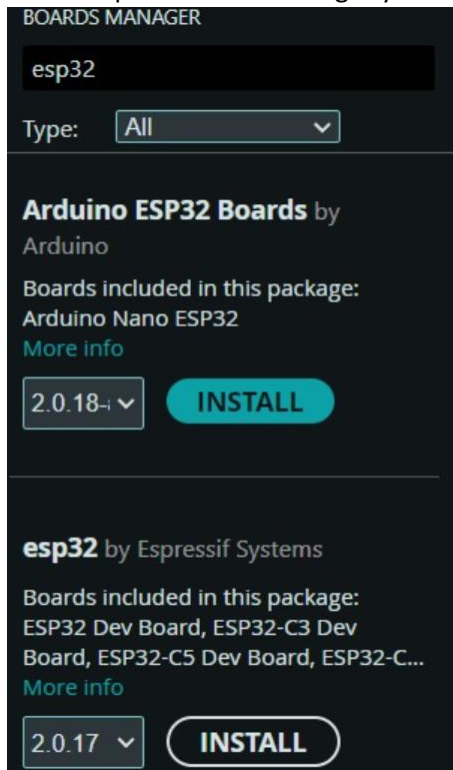
Misal masih tidak bisa, install app berikut :

<https://drive.google.com/file/d/1mjJGaasvh0iiqljNrpCc17mMRMeVIDaC/view>

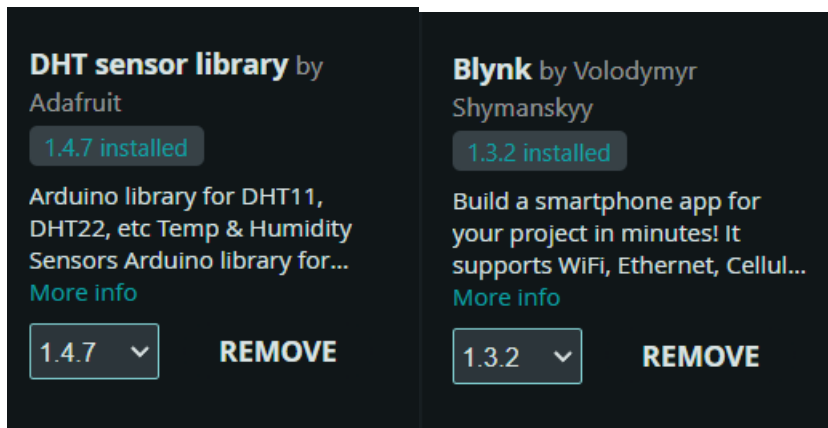
7. Pilih Board ESP32 Dev Module



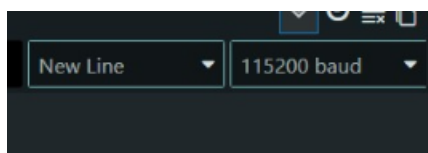
8. Install pada boards manager yaitu esp32 by Espressif Systems versi 2.0.17



9. Buka library lalu install **DHT Sensor Library by Adafruit** dan **Blynk by Volodymyr Shymanskyy**



10. Ubah pada baud pada serial monitor menjad 115200 baud



11. Lalu klik upload sehingga output seperti ini

```
Output Serial Monitor
Writing at 0x00000000... (92 %)
Writing at 0x000f2d75... (94 %)
Writing at 0x000f8259... (97 %)
Writing at 0x000fd8f8... (100 %)
Wrote 989632 bytes (634050 compressed) at 0x00010000 in 56.1 seconds (effective 141.2 kbit/s)...
Hash of data verified.

Leaving...
Hard resetting via RTS pin...
```

BLYNK IOT Laptop

1. Buka website blynk.io lalu buat akun

2. Buat template dengan nama “Monitoring DHT 11” dengan hardware ESP32 dan connection type WiFi

Create New Template

NAME
Monitoring DHT 11 19 / 50

HARDWARE CONNECTION TYPE
ESP32 WiFi

DESCRIPTION
Description 0 / 128

Cancel Done

3. Copy Firmware configuration pada bagian home, lalu paste pada app Arduino IDE pada bagian DHT11_ESP32.ino (pastikan copy paste di tempat yang sudah disediakan)

```
Firmware configuration
Template ID and Template Name should
be declared at the very top of the
firmware code.

#define BLYNK_TEMPLATE_ID
"TMPL6oSFfx5ra"
#define BLYNK_TEMPLATE_NAME
"Monitoring DHT 11 2"
```

```
// 1. DEFINISI BLYNK (Wajib di urutan paling atas)
#define BLYNK_TEMPLATE_ID "TMPL6oSFfx5ra"
#define BLYNK_TEMPLATE_NAME "Monitoring DHT 11 2"
#define BLYNK_DEVICE_NAME "Monitoring DHT 11"
#define BLYNK_FIRMWARE_VERSION "0.1.0"
```

3. Lalu pilih ke bagian Datastreams dan buat baru pilih “Virtual Pin” untuk Suhu dan buat seperti ini

Virtual Pin Datastream

General Expose to Automations

 NAME

Alias



PIN

DATA TYPE

UNITS

MIN

MAX

DECIMALS

DEFAULT VALUE

☐ Enable history data

Cancel

Create

4. Kemudian, buat virtual pin yang kedua untuk Kelembapan dan buat seperti ini

Virtual Pin Datastream

General Expose to Automations

 NAME

Alias



PIN

DATA TYPE

UNITS

MIN

MAX

DECIMALS

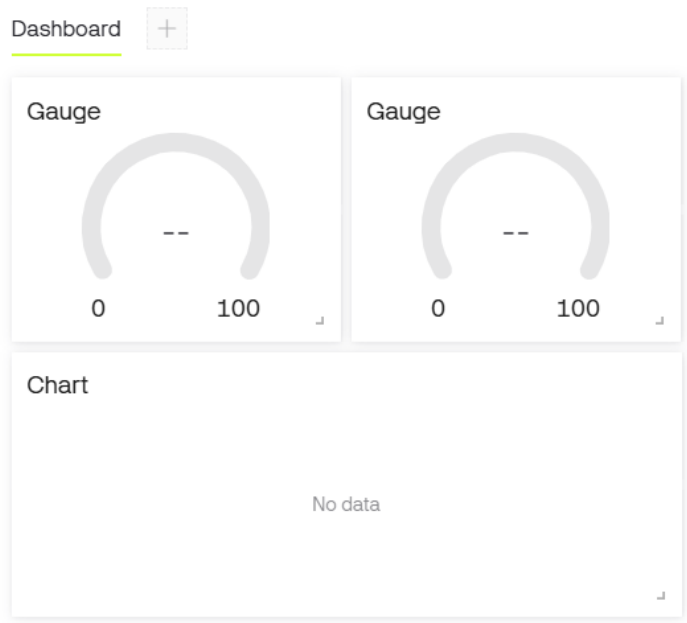
DEFAULT VALUE

☐ Enable history data

Cancel

Create

5. Pergi ke bagian Web Dashboard lalu buat seperti ini



6. Pada setiap Gauge, pilih masing2 data stream suhu dan kelembapan

Gauge Settings

TITLE (OPTIONAL)

Gauge

Datastream

Choose Datastream



or

+ Create Datastream

Kelembapan (V1)

Suhu (V0)

7. Pada bagian chart, pilih kedua datastream

Chart Settings

TITLE (OPTIONAL)

Chart

Datastreams

You have no datastreams to select

+ Create Datastream



Kelembapan (V1)

▼

AVG of

▼





CHART TYPE

COLOR



Y-AXIS

☐ Show Y-axis

☐ Autoscale

☐ Override Datastream's Min/Max fields

MINIMUM Y AXIS RANGE

Minimum Y axis range



Suhu (V0)

▼

AVG of

▼

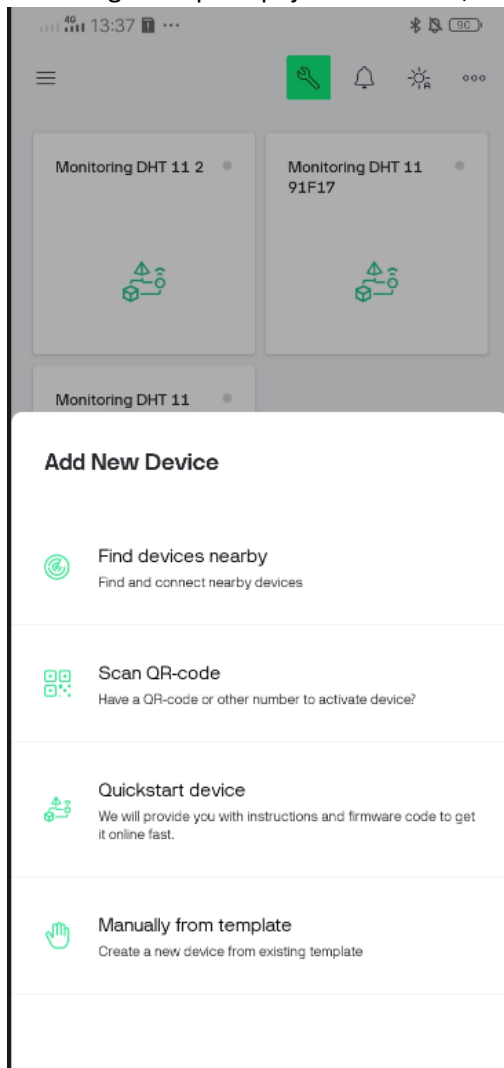




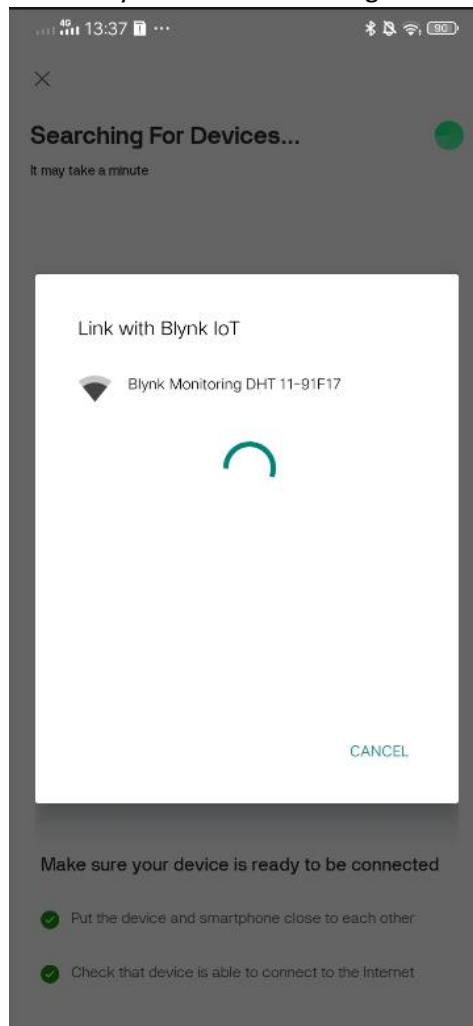
BLYNK IOT Handphone

1. Login akun yang sama dengan Blynk IoT pada laptop

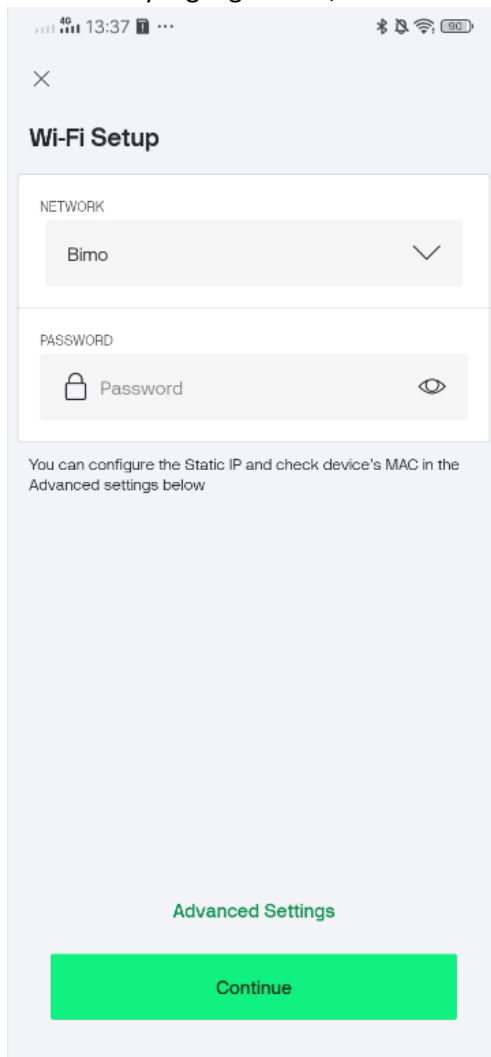
2. Klik tiga titik pada pojok kanan atas, lalu pilih “Find devices nearby



3. Pilih Blynk IoT device masing2



4. Pilih wifi yang digunakan, lalu masukkan passwordnya (pastikan wifi laptop dan handphone sama)



The image shows a smartphone screen with a 'Wi-Fi Setup' interface. At the top, there's a status bar with '4G', signal strength, time '13:37', and battery level '100'. Below the status bar is a close button (X). The title 'Wi-Fi Setup' is prominently displayed. The main content area is divided into two sections: 'NETWORK' and 'PASSWORD'. The 'NETWORK' section has a dropdown menu currently showing 'Bimo'. The 'PASSWORD' section has a text input field with a lock icon on the left and an eye icon on the right, with the placeholder text 'Password'. Below these sections, a note states: 'You can configure the Static IP and check device's MAC in the Advanced settings below'. At the bottom, there is a link for 'Advanced Settings' and a large green 'Continue' button.

13:37 100

Wi-Fi Setup

NETWORK

Bimo

PASSWORD

Password

You can configure the Static IP and check device's MAC in the Advanced settings below

Advanced Settings

Continue

5. Jika sudah terdapat angka suhu dan kelembapannya, maka IoT sudah bisa digunakan

